

No. of Functional Features Included in the Solution

Date	2 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Functional Features Overview:

This document presents the functional features implemented in the **Voice-Based Concept Understanding Analyser (VBCUA)**. Each feature has been designed to support the automated evaluation of spoken conceptual explanations by combining artificial intelligence, natural language processing, and audio signal processing techniques. The system enables users to upload audio, transcribe speech into text, analyze semantic similarity with reference concepts, extract speech fluency metrics, generate intelligent comprehension scores, visualize waveforms, and download detailed PDF reports. The implementation status of each feature is documented to demonstrate the completeness of the solution and its contribution toward achieving the project's objectives.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Upload	Allows users to upload WAV/MP3 audio files for evaluation.	Streamlit UI	Done	0.5
2	Speech-to-Text Transcription	Converts spoken audio into text using OpenAI Whisper.	speech_to_text.py	Done	1.0
3	Semantic Similarity Analysis	Compares transcribed text with reference content using Sentence-BERT.	semantic_eval.py	Done	1.0
4	Audio Feature Extraction	Extracts speech features such as RMS energy, pauses, and duration using Librosa.	audio_features.py	Done	0.5
5	Intelligent Scoring & Feedback	Calculates conceptual understanding score and generates qualitative feedback.	score_calculator.py, feedback_generator.py	Done	0.5
6	Waveform Visualization	Generates waveform image for uploaded audio.	waveform_generator.py	Done	0.5
7	PDF Report Generation	Creates downloadable evaluation reports with scores and analysis.	report_generator.py	Done	0.5
8	Interactive Dashboard	Displays transcription, similarity score, waveform, feedback, and report download.	app.py	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Audio Upload, Dashboard, PDF Download
2	Backend / Logic	3	Speech-to-Text, Semantic Analysis, Intelligent Scoring
3	Database / Storage	1	Local Storage of Audio Files, Waveforms, Reports
4	API / Integration	2	OpenAI Whisper, Sentence-BERT
5	Security / Authentication	0	Not Implemented (Future Enhancement)